

Fig. 5.3 Two examples in which we suppose the accessible domain at given energy is the annulus. On the *left*, periodic transport of a collection of points with positive area is evidence of non-ergodic motion (the union of the four disks shown would constitute an invariant region of positive measure). In the figure at *right* a quasi-periodic space-filling trajectory is shown. The oscillatory motion, although quite regular in appearance, gradually fills in the whole domain

where Ω_1, Ω_2 are two nonzero parameters. Because the two systems involving q_1, p_1 and q_2, p_2 are not coupled in the equations of motion, this system has two invariants $H_1 = p_1^2/2 + \Omega_1^2 q_1^2/2$ and $H_2 = p_2^2/2 + \Omega_2^2 q_2^2/2$. For any $\mathrm{E}_1, \mathrm{E}_2 > 0$, the set of points satisfying $H_1 = \mathrm{E}_1$, $H_2 = \mathrm{E}_2$ is a two-dimensional torus. If the frequencies Ω_1 and Ω_2 are not rationally related (i.e. Ω_1/Ω_2 is an irrational number), then for any $\mathrm{E}_1, \mathrm{E}_2$ the torus will eventually be filled in by any trajectory. On the other hand, the energy surface $H = \mathrm{E}$ is the union of all such 2-tori with $\mathrm{E}_1 + \mathrm{E}_2 = \mathrm{E}$. There is no mechanism for the system to move from one torus to the other, so the system is not ergodic.

In typical molecular dynamics applications with multiple bodies and complicated force laws, the motion is assumed to be chaotic. Ergodic coverage of Ω_E would then have to arise from the global properties of a chaotic system (sensitivity to initial conditions and transitivity). Liouville's equation,

$$\frac{\partial \rho}{\partial t} = \mathcal{L}_f^\dagger \rho,$$

describes how a distribution of initial conditions is carried forward under the time evolution of a dynamical system. That is, starting from the initial distribution of phase space points described by $\rho(z, 0)$, we arrive at a new distribution $\rho(z, t)$. We will see in an example to be presented below that various sets of initial conditions with uniform energy E may appear to spread out to cover (become dense in) a particular region, the accessible energy surface at energy level E.

An even stronger property than the ergodic property is the concept of a *mixing* system. For a mixing system, the finite time density $\rho(z, t)$ converges, in the weak sense, to the invariant distribution $\rho_\infty(z)$, as $t \to \infty$. That is, we have, for all test functions ϕ in some chosen space

$$\lim_{t\to\infty} \langle \phi, \rho(\cdot, t) \rangle = \langle \phi, \rho_\infty \rangle.$$